

Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology/ Department of Computer Applications		
Department Name	Department of Computer Applications		
Programme Name	Bachelor of Computer Applications (AI & ML)		
Course Name	Programming for AI	Session	July-Dec 2025
Course Code	25COA1102	Semester/Batch	Ist
L-T-P(Per Week)	2-0-4	Course Credits	4
Pre-requisite	NA	NHEQF Level ¹	4,5
Course Coordinator	Dr. Mohd Shuaib Khan	SDG Number ⁴	4,9

1. Objectives of the Course

Python is a general-purpose language that can be used for developing desktop application, web applications and microcontroller-based applications. This course primarily focuses on teaching core python concepts. After the course completion the students will be able to make GUI based applications for real world problems.

The main objectives of the course are:

- Exploring python popularity as a general-purpose programming language.
- Understanding python's way to deal with numbers and strings.
- Working with python internal data structures.
- Diving into python modular programming style.
- Exploring object-oriented nature of python programming.
- Understanding and developing GUI applications.

2. Course Learning Outcomes (CLOs)

Student should be able to:

	CLOs	Program Outcomes (PO)	NHEQF Level Descriptor ²	No. of Lectures
CLO01	To understand the python fundamentals and environment.	PO1,PO4,PO8	Q1, Q2	20
CLO02	To learn the various ways to perform operations on numbers and strings.	PO1, PO4, PO8	Q3, Q6	20
CLO03	Logic building using looping and decision statements.	PO3	Q3	24
CLO04	Study internal data structures like list, tuples, sets and dictionaries.	PO2, PO6	Q1, Q2	20
CLO05	Learn Professional way of writing code using modular and object-oriented approach.	PO1, PO4, PO8	Q3	20
Total Contact Hours				104

¹ National Higher Education Qualification Framework Level, Refer to annexure

² NHEQF Level Descriptor, Refer to Annexure & [Learning outcomes descriptors for qualification for all levels on the NHEQF](#)

³Types of Assessments can be referred from Type of Assessments. Refer to Annexure.

⁴For SDG Mapping with Courses, PI refer [SDG Mapping policy for Courses](#)

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	Type of Assessment's ³
CLO01	H			H				M						FA/SA
CLO02	H			M				H						FA/SA
CLO03			M											FA/SA
CLO04		M				L								FA/SA
CLO05	L			H				L						FA/SA

H=High, M=Medium, L=Low

3. Recommended Books:**B01:** Lutz, Mark, Learning python; 2013; O'Reilly Media, Inc**B02:** Barry, Paul; Head first python; 2010; O'Reilly Media, Inc**B03:** Swaroop, C; A Byte of Python; 2013; Webshel Inc**B04:** Ravichandran, R.Saravanan, "Learn to Master Python", Staredu Solution, c2018.**B05:** Roberto Tamassia, Michael H Goldwasser, Michael T Goodrich, "Data Structures and Algorithms in Python", 1st Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978-81265621764. Other readings and relevant websites:

Serial No	Link of Journals, Magazines, websites and Research Papers
1.	http://www.python.org/
2.	https://learnpython.org/
3.	https://pythoninstitute.org/
4.	https://www.javatpoint.com/python-tutorial
5.	https://www.kaggle.com/learn/python

5. Recommended Tools and Platforms

Python 3 IDLE, Anaconda 3 (Jupyter notebook 3.6.0), Google colab, Replit

6. Course Plan: Theory+ LabPlan Theory Plan

Lect. No.	Topic(s)
1-3	Introduction to Python programming: Python overview, features, Python environment setup, lines and indentation, comments, statements, print() usage.
4-8	Numbers, Strings, and Math Library: Variables, number types, math functions (pow, sqrt, ceil, etc.), string operations and methods, string formatting, number conversion
9-15	Control Flow and Logic Building: if, if-else, elif, nested conditions, while, for loops, flow control, loop-based logic development
	ST1 (Syllabus=30% (Lect. No. 1-15))
16-19	Data Structures: Lists, sets and dictionaries: creation, access, methods, slicing, comprehension, nesting, deletion, updates, and input methods
20-22	Tuples: Set and Dictionaries- Introduction, methods, comprehension and its relative properties
23-25	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces ,Reloading, Built in modules, Generating Random values.
	ST2 (Syllabus=60% (Lect. No. 16-25))
26-30	Object-Oriented Programming : Functions, recursion, local/global scope, parameter passing (*args, **kwargs), classes, objects, constructors, inheritance, polymorphism
31-36	GUI Programming and Applications: Tkinter, PyQt installation, widgets (buttons, textboxes, etc.), event handling, signals/slots, menus, calendar, mini projects (e.g., game, calendar app)
	End Term Exam (Syllabus=100% (Lect No. 1-36))

Lab Plan

Lab No.	Topic(s)
1-2	Create a new file named 'hello_tools.ipynb'. Print 'Hello, Python!', check Python version using sys.version, and create a simple addition program that takes two inputs from the user and prints the sum.
3-4	1. Write a python program for printing a statement with 'sep' parameter. 2. Write a python program for printing a statement with 'end' parameter.
5-6	Write a Python program to convert temperatures to and from Celsius, Fahrenheit. [Formula: $c/5 = f-32/9$].
7-8	Draw a table to show first 20 natural numbers and their equivalent hexadecimal, octal and binary values
9-10	Write a program to find the factorial of a number.
11-12	Write a program that finds whether a given character is present in a given string or not. In case it is present, then print the index at which it is present. Do not use built in find function to search the character
13-14	Write a python program to define a module to find Fibonacci Numbers and import the module to another program.
15-16	Write a code to having a method that can receive positional parameters, keyword arguments as well as * and ** based arguments at a same time and process them in meaningful way.
17-18	Generate a student-name list of 15 students and then accepts two values from user namely startValue and endValue and perform slicing on the list. Finally count the length of every item in sliced list and display it.
19-20	Write a program to append the elements of the list.
Continuous Evaluation 1	
21-22	Generate a list dynamically named "Raw_List" having 10 elements and then create two another lists, one to store squares of every number present in "Raw_List" and second to store factorial of every number present in "Raw_List".
23-24	Create dynamic dictionary for 10 students. The data should be received from keyboard. The key should be roll number and value should be name, age, stream and contact number. Finally search a particular roll number and display its corresponding information.
25-26	Generate a table having two columns. First column should display keys and second column should display corresponding values. Initial dictionary should have at-least 10 records. At the end of table show total number of records available in dictionary.
27-28	Write a code to employ inheritance in python and take dictionaries as data member in classes
29-30	Understanding and Installing PyQt4 and QT Designer Interface
31-32	Create a GUI with various controls and menu components.
33-34	Using GUI controls (push-buttons, text boxes, radio buttons, checkboxes, labels, managing resource files, combo boxes, list boxes, menus and sub menu's, calendar control and other controls), design python program to demonstrate GUI controls
Continuous Evaluation 2	

7. Delivery/Instructional Resources
Theory Plan:

Lect. No.	Topics	CLO	Book No, CH No, Page No	TLM ³	ALM ⁴	Web References	Audio-Video
1-3	Introduction to Python programming: Python overview, features, Python environment setup, lines and indentation, comments, statements, print() usage.	CLO01	B01, CH 1, Page no 1-17 B02, CH 2, Page No 19-24 B03, CH 03, Page No 83-124	Lecture	Quiz, Test questions	https://www.anaconda.com/ https://www.learnpython.org/ https://docs.python.org/3/tutorial/introduction.html	https://nptel.ac.in/courses/106106182
4-8	Numbers, Strings, and Math Library: Variables, number types, math functions (pow, sqrt, ceil, etc.), string operations and methods, string formatting, number conversion	CL002	B02, CH 6, Page No 117-149 B03, CH 04, Page No 137-179	Lecture	Quiz, Puzzles, Tests	https://docs.python.org/3/tutorial/controlflow.html	https://nptel.ac.in/courses/106106182
9-15	Control Flow and Logic Building: if, if-else, elif, nested conditions, while, for loops, flow control, loop-based logic development	CL002	CLO02 B02, CH 6, Page No 117-149	Lecture	Quiz, Brain Storming Sessions	https://docs.python.org/3/tutorial/controlflow.html https://docs.python.org/3/tutorial/controlflow.html#for-statements	https://nptel.ac.in/courses/106106182 https://nptel.ac.in/courses/106106182
16-19	Data Structures: Lists, sets and dictionaries: creation, access, methods, slicing, comprehension, nesting, deletion, updates, and input methods	CL003	B01, CH 4, Page No 81-98 B02, CH 9, Page No 237-280	Lecture	Quiz, Peer Review	https://docs.python.org/3/tutorial/controlflow.html#more-on-defining-functions	https://www.coursera.org/lecture/python-genomics/lecture-5-1-functions-part-1-5-54-PGXxZ https://www.coursera.org/lecture/python-genomics/lecture-5-2-functions-part-2-8-20-ylbSK

³ Teaching Learning Methods, Refer to Annexure

⁴ Active Learning Methods

20-22	Tuples: Set and Dictionaries- Introduction, methods, comprehension and its relative properties	CL004	B02, CH 10, Page No 283-306	Lecture	Test Questions, Quiz	https://docs.python.org/3/tutorial/datastructures.html	https://nptel.ac.in/courses/106106182
23-25	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces, Reloading, Built in modules, Generating Random values.	CL004	B02, CH 10, Page No 283-306 B04, CH 08, Page No 238	Lecture	Test Questions	https://docs.python.org/3/tutorial/datastructures.html#list-comprehensions	https://nptel.ac.in/courses/106106182
26-30	Object-Oriented Programming : Functions, recursion, local/global scope, parameter passing (*args, **kwargs), classes, objects, constructors, inheritance, polymorphism	CL004	B02, CH 8, Page No 207-235	Lecture	Test Questions	https://docs.python.org/3/tutorial/controlflow.html#documentation-strings	https://www.coursera.org/lecture/python-representation/python-strings-bbBTs https://www.coursera.org/lecture/python-data-analysis/python-more-on-strings
31-36	GUI Programming and Applications: Tkinter, PyQt installation, widgets (buttons, textboxes, etc.), event handling, signals/slots, menus, calendar, mini projects (e.g., game, calendar app)	CLO05	B01, CH 5, Page No 101-111	Lecture	Brain Storming Sessions, Focused Listing	https://python-course.eu/python-tutorial/file-management.php	https://www.coursera.org/lecture/python-genomics/lecture-6-modules-and-packages-10-32-rejwK

Lab Plan:

Lab No.	Experiment	CLO	TLM	ALM	Web References	Audio-Video
1-2	Create a new file named 'hello_tools.ipynb'. Print 'Hello, Python!', check Python version using sys.version, and create a simple addition program that takes two inputs from the user and prints the sum.	CLO01	Lecture	Think / Pair / Share	https://www.geeksforgeeks.org/introduction-to-python/	https://www.youtube.com/watch?v=xEK_eIMf37A
3-4	1. Write a python program for printing a statement with 'sep' parameter. 2. Write a python program for printing a statement with 'end' parameter.	CLO01	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/python-data-types/	https://www.youtube.com/watch?v=0VMGSADIs4M
5-6	Write a Python program to convert temperatures to and from Celsius, Fahrenheit. [Formula: $c/5 = f-32/9$].	CLO02	Active Learning	Think / Pair / Share	https://www.javatpoint.com/control-statements-in-python	https://www.youtube.com/watch?v=9xiFcK3MRYA
	Draw a table to show first	CLO0	Active	Think / Pair /	https://www.geeksforge	https://www.youtub

7-8	20 natural numbers and their equivalent hexadecimal, octal and binary values	1	Learning	Share	eks.org/python-string-methods-set-1-find-rfind-startwith-endwith-islower-isupper-lower-upper-swapcase-title/	e.com/watch?v=ErgSKwpEEDs
9-10	Write a program to find the factorial of a number.	CLO04	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/g-fact-41-multiple-return-values-in-python/	https://www.youtube.com/watch?v=jpqi_HRdpGA
11-12	Write a program that finds whether a given character is present in a given string or not. In case it is present, then print the index at which it is present. Do not use built in find function to search the character	CLO04	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/keyword-and-positional-argument-in-python/	https://www.youtube.com/watch?v=kB829ciAXo4
13-14	Write a python program to define a module to find Fibonacci Numbers and import the module to another program	CLO03	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/python-lists/	https://www.youtube.com/watch?v=Icw4cLTtKh4
15-16	Write a code to having a method that can receive positional parameters, keyword arguments as well as * and ** based arguments at a same time and process them in meaningful way	CLO03	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/list-methods-python/	https://www.youtube.com/watch?v=DZliigDWF_M

17-18	Generate a student-name list of 15 students and then accepts two values from user namely startValue and endValue and perform slicing on the list. Finally count the length of every item in sliced list and display it.	CLO03	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/initialize-an-empty-dictionary-in-python/	https://www.youtube.com/watch?v=brnsFCa7npc
19-20	Write a program to append the elements of the list	CLO03	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/python-dictionary/	https://www.youtube.com/watch?v=brnsFCa7npc
21-22	Generate a list dynamically named "Raw_List" having 10 elements and then create two another lists, one to store squares of every number present in "Raw_List" and second to store factorial of every number present in "Raw_List".	CLO04	Active Learning	Think / Pair / Share	https://www.w3schools.com/python/python_cl asses.asp	https://www.youtube.com/watch?v=DBWde9iWBLw
23-24	Create dynamic dictionary for 10 students. The data should be received from keyboard. The key should be roll number and value should be name, age,	CLO04	Active Learning	Think / Pair / Share	https://www.w3schools.com/python/python_cl asses.asp	https://www.youtube.com/watch?v=DBWde9iWBLw

	stream and contact number. Finally search a particular roll number and display its corresponding information					
25-26	Generate a table having two columns. First column should display keys and second column should display corresponding values. Initial dictionary should have at-least 10 records. At the end of table show total number of records available in dictionary	CLO05	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/python-designing-gui-applications-using-pyqt/	https://www.youtube.com/watch?v=nzzQvRISyUY
27-28	Write a code to employ inheritance in python and take dictionaries as data member in classes	CLO05	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/python-designing-gui-applications-using-pyqt/	https://www.youtube.com/watch?v=Dmo8eZG5I2w
29-30	Understanding and Installing PyQt4 and QT Designer Interface	CLO01	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/python-data-types/	https://www.youtube.com/watch?v=0VMGSADIs4M
31-32	Create a GUI with various controls and menu components	CLO02	Active Learning	Think / Pair / Share	https://www.javatpoint.com/control-statements-in-python	https://www.youtube.com/watch?v=9xiFcK3MRYA
33-34	Using GUI controls (push-buttons, text boxes, radio buttons,	CLO01	Active Learning	Think / Pair / Share	https://www.geeksforgeeks.org/python-string-methods-set-1-find-rfind-startwith-endwith-islower-isupper-lower-upper-	https://www.youtube.com/watch?v=ErgSKwpEEDs

	checkboxes, labels, managing resource files, combo boxes, list boxes, menus and sub menu's, calendar control and other controls), design python program to demonstrate GUI controls				swapcase-title/	
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8. Remedial Classes⁵

After every Sessional Test, different types of learners will be identified and special discussions will be planned and scheduled accordingly.

9. Self-Learning⁶

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Topics	CLO	ALM	References/MOOCs
1	Installation, Conditional Statements, Functions, Loops and Iterations	CLO01, CLO02, CLO05	Think – Pair-Share, Peer Review	Courseera-Python for Everybody Specialization (Getting Started with Python)
2	Data Types in Python, Loops, Strings, List	CLO01, CLO02, CLO04	Think – Pair-Share, Brain Storming sessions	Courseera- Crash Course on Python

10. Delivery Details of Content Beyond Syllabus⁷

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCs
1	Fundamentals of python library: Numpy :	CLO02, CLO03	PO1, PO2, PO3, PO4, PO5, PO10	Think – Pair- Share, Peer Review	https://numpy.org/doc/stable/user/basics.html https://numpy.org/doc/stable/user/absolute_beginners.html

⁵ Refer to Annexure

⁶ Refer to Annexure

⁷ Refer to Annexure

	Array creation, Adding ,removing, and sorting elements of array, array dimension, reshaping, 1D, 2D arrays				
2	Fundamentals of python library: Pandas	CLO02, CLO03, CLO04	PO1,PO2,PO3,PO4, PO5, PO10	Think – Pair- Share, Peer Review, Brain Storming	https://pandas.pydata.org/docs/user_guide/index.html

11. Evaluation Scheme & Components:

Assessment Type ⁸	Evaluation Component ⁹	Type of Component ¹⁰	No. of Assessments ¹¹	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component1	Continuous Lab Evaluations		20%	20	Offline	CLO01- CLO06
		Lab Performance	02*	Viva:40% File:30%	10		
		Internal Viva	01*	Written/Execution: 30%	10		
Summative	Component2	Sessional Tests(STs)	02**	30%	30	Offline	CLO01- CLO06
Summative	Component3	End Term Examination	01***	50%	50	Offline	CLO01- CLO06

*Lab performance and Internal Viva is a mandatory evaluation taken during the semester. It is considered based on Lab practical file, viva and written/execution. Two Lab Performances(LP) and one Internal Viva (IV) is to be taken during the semester and the ERP system will automatically pick best of two lab performances and marks of internal viva for awarding final lab internal marks.

**Out of 2 Sessional Test (ST) the ERP system will automatically pick best of 2 ST's for awarding marks. ST's to be held as per departmental academic calendar.

*** As per academic guidelines minimum 75% attendance is required to become eligible for appearing in the End Semester Examination.

⁸ Refer to [Annexure 2 of NCrF](#)

⁹ Refer to Annexure

¹⁰ Refer to Annexure

¹¹ Refer to Annexure

12. Syllabus of the Course:

Subject:			
S.No.	Topic(s)	No. of Lectures	Weightage %
1	Introduction to Python programming: Python overview, features, Python environment setup, lines and indentation, comments, statements, print() usage	3	5%
2	Numbers, Strings, and Math Library: Variables, number types, math functions (pow, sqrt, ceil, etc.), string operations and methods, string formatting, number conversion	4	10%
3	Control Flow and Logic Building: if, if-else, elif, nested conditions, while, for loops, flow control, loop-based logic development	7	20%
4	Data Structures: Lists, sets and dictionaries: creation, access, methods, slicing, comprehension, nesting, deletion, updates, and input methods	5	15%
5	Tuples: Set and Dictionaries- Introduction, methods, comprehension and its relative properties	5	15%
6	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces, Reloading, Built in modules, Generating Random values.	5	15%
7	Object-Oriented Programming : Functions, recursion, local/global scope, parameter passing (*args, **kwargs), classes, objects, constructors, inheritance, polymorphism	7	20%

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Mohd Shuaib Khan	
Program Incharge	Dr. Maninderjit Singh Khanna	
Pro Vice Chancellor	Dr. Jaiteg Singh	
Date(DD/MM/YYYY)	15/09/2025	